

William Ronchetti

Senior Backend / Infrastructure Engineer

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7+ years building secure, scalable AWS data platforms in Python — infrastructure-as-code, LLM/RAG systems, and FISMA-Moderate, HIPAA & GDPR-compliant delivery.

TECHNICAL SKILLS

Languages: Python, SQL, C/C++

Backend & Data: PostgreSQL (RDS), Elasticsearch / Lucene, Redis, DynamoDB, Amazon Athena, S3, Parquet, Kafka, SQS; Pyramid, Flask, boto3, SQLAlchemy, pytest

AI / LLM: AWS Bedrock, retrieval-augmented generation (RAG), vector search, text-to-SQL, embeddings, PHI de-identification

Infrastructure & IaC: AWS CloudFormation (Troposphere), Terraform, VPC, EC2, ECS (Fargate), ECR, Lambda, Batch, Step Functions, API Gateway, Route 53, Elastic Beanstalk; Docker, Nginx, Apache

Observability & CI/CD: CloudWatch, Splunk (HEC), Fluent Bit / FireLens, Sentry, GitHub Actions, Travis CI

Security & Compliance: AWS KMS, Secrets Manager, Security Hub, Certificate Manager (ACM); FISMA-Moderate, HIPAA, GDPR

Scientific / Pipelines: Nextflow, Seqera

PROFESSIONAL EXPERIENCE

Novo Nordisk (contract) — Data Product Engineer

Remote • Sept 2024 – Present

- Lead backend engineer for the **Omics Platform**, an internal AWS system managing petabytes of diverse omics data used by dozens of research teams for cataloging and analysis.
- Built an **LLM-based document intelligence system** that ingests thousands of scientific PDF reports (born-digital and scanned), extracts structured data, and serves a chat agent over a hybrid retrieval architecture combining vector search and text-to-SQL on AWS Bedrock.
- Designed a two-zone **PHI de-identification pipeline** to meet HIPAA requirements, enabling LLM processing of sensitive documents with small-cell suppression and controlled re-identification.
- Built a highly scalable ingestion pipeline on AWS Lambda, Glue, and Batch for validation and transfer of raw data and discovery metadata, processing hundreds of datasets across several dozen vendors/CROs.
- Designed a Parquet/Athena metadata model for tracking diverse omics datasets, reducing dataset onboarding and discovery time for research teams.
- Review code and mentor a small team of engineers on software, cloud-engineering, and security best practices to ensure GDPR compliance.

Harvard Medical School — Biomedical Informatics — Senior Software Engineer

Boston, MA • Aug 2019 – Present

- Lead backend development in Python for the **CGAP, 4DN, and SMaHT** data platforms, supporting petabytes of raw data and associated metadata.
- Refactored complex, organically grown AWS infrastructure into reproducible **infrastructure-as-code (CloudFormation)**, achieving FISMA-Moderate compliance and passing multiple independent security audits with workflows for promptly resolving vulnerabilities.
- Migrated the web platform from Apache on Elastic Beanstalk to **Nginx on ECS (Fargate)**, cutting web-portal compute costs by 50% and enabling traffic-based autoscaling.
- Implemented observability with structured event logging via a **Splunk HEC** pipeline and container log shipping through FireLens / Fluent Bit.
- Optimized Postgres and Elasticsearch query performance by resolving aggregation timeouts caused by large terms filters, cutting response time by 50%.
- Devised and implemented advanced security protocols (encryption, secrets management, access controls) and mentored junior engineers and domain experts on feature development and code review.

EDUCATION

Cornell University — Ithaca, NY

M.Eng., Computer Science (Dec 2018) • B.A., Computer Science (May 2018). Varsity Cross Country, Indoor & Outdoor Track. Associate in Research (part-time), Duke University Dept. of ECE — C/C++ course assignments on Coursera (2019–present). Teaching Assistant, Operating Systems & System Security (Cornell).

PUBLICATIONS

Co-author on 5 peer-reviewed publications (267 citations), including 2 in *Nature* and 2 in *Nature Communications* (2022–2025).